

AI-Based Quality Risk Prediction Report

Date: 20-07-2026

Time: 01:08 PM

Prediction Summary

Prediction	FAIL
Failure Risk	57.66%
Risk Level	Moderate Risk
Likely Defect	Wall Thickness Variation
Confidence	96.41%

Input Parameters

Parameter	Value
Material	PU
Braid	None
Machine	EXT-01
Target OD (mm)	20.0
Target ID (mm)	17.5
Wall Thickness (mm)	1.25
OD Tolerance (mm)	0.2
ID Tolerance (mm)	0.2
Wall Tolerance (mm)	0.08
Vacuum (bar)	-0.8094
Screw Speed (RPM)	78.41
Zone 2 Temp (°C)	192.34
Zone 3 Temp (°C)	205.28
Melt Pressure (bar)	227.05
Puller Speed (m/min)	10.297

Material Moisture (%)	0.0661
-----------------------	--------

AI Recommendations

- Increase Melt Pressure from 227.05 to 238.40 bar
- Decrease Screw Speed from 78.41 to 74.49 RPM
- Increase Puller Speed from 10.30 to 10.81 m/min
- Decrease Material Moisture from 0.07 to 0.06 %
- Increase Vacuum from -0.81 to -0.77 bar

Generated by: AI-Based Thermoplastic Extrusion Quality Prediction System